Course: Computer Vision

Course Code: ITA0506

Slot: D

Faculty: Jeni Gracia R C

Submitted By

Y. Sravan Kumar (192472289)

# CO4_AT3_COMPARATIVE ANALYSIS

## 1. Precision and Recall

### Model A — Viola-Jones

Given:

TP = 80

FP = 20

FN = 30

**Precision:**

Precision=TP+FPTP =80+2080 =10080 =0.80=80%

**Recall:**

Recall=TP+FNTP =80+3080 =11080 =0.727=72.73%

### Model B — YOLO

Given:

TP = 90

FP = 15

FN = 20

**Precision:**

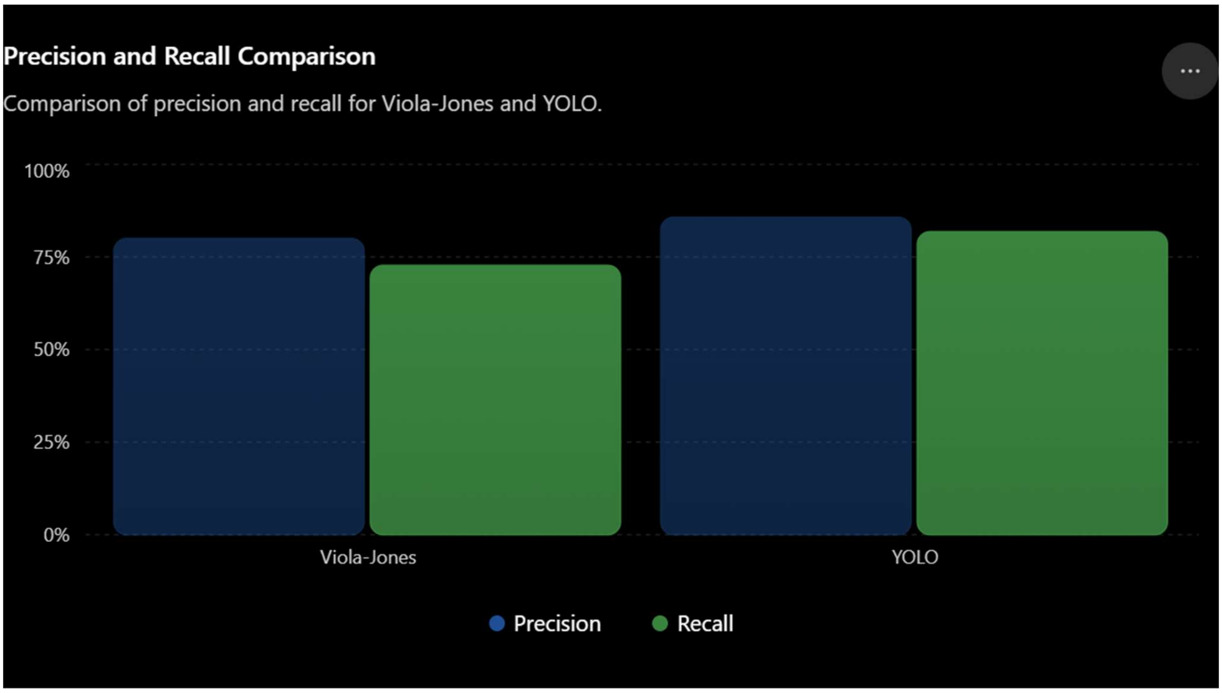
Precision=90+1590 =10590 =0.857=85.71%

**Recall:**

Recall=90+2090 =11090 =0.818=81.82%

## Comparison

| Model | Precision | Recall |
|---|---|---|
| Viola-Jones | 80% | 72.73% |
| YOLO | 85.71% | 81.82% |




## Final Answer

YOLO provides better overall object detection performance because it has both higher precision (85.71%) and higher recall (81.82%) than Viola-Jones.

Higher precision → fewer false detections.

Higher recall → fewer missed objects.

Therefore, Model B (YOLO) is the better model.

## 2. Motion Estimation Error Analysis

Given

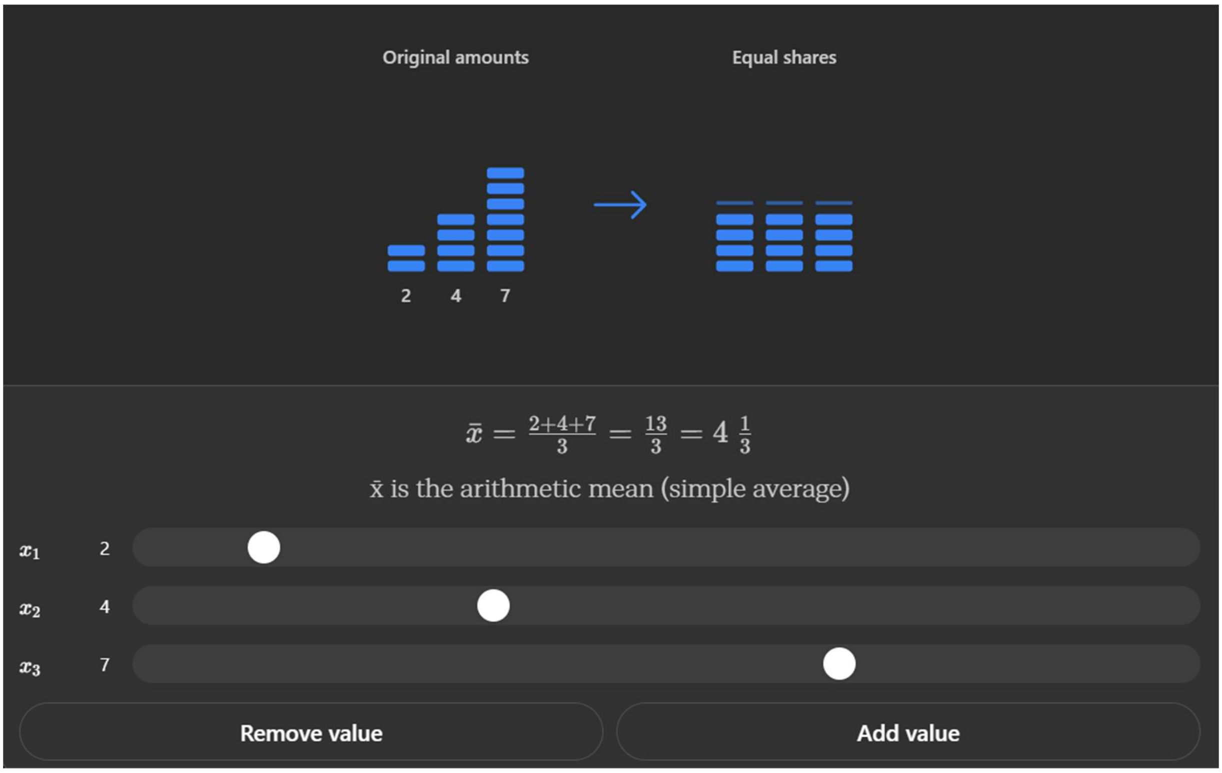
Method A: 2, 3, 2 pixels

Method B: 3, 5, 4 pixels

### (a) Average Displacement Error

Method A:

Average Error=32+3+2 =37 =2.33 pixels




Method B:

Average Error=33+5+4 =312 =4 pixels

### (b) Comparison

| Method | Average Error |
|---|---|
| Method A | 2.33 pixels |
| Method B | 4 pixels |

Method A is more accurate because it has a lower average displacement error.

## (c) Justification

Lower displacement error means the estimated motion is closer to the actual motion. Method A has an average error of 2.33 pixels, while Method B has 4 pixels.

Therefore, Method A provides more accurate motion estimation and is the better choice.

# 3. Depth Estimation Comparison

Given

| Object | Actual Depth | Stereo Vision | SfM |
|---|---|---|---|
| 1 | 2.0 m | 2.0 m | 2.5 m |
| 2 | 3.0 m | 3.0 m | 3.8 m |
| 3 | 4.0 m | 4.0 m | 5.0 m |

## (a) Absolute Error

Formula:

$\text{Absolute Error} = |\text{Estimated Depth} - \text{Actual Depth}|$

## Stereo Vision

Object 1:

$|2.0-2.0|=0$ m

Object 2:

$|3.0-3.0|=0$ m

Object 3:

$|4.0-4.0|=0$ m

## SfM

Object 1:

$|2.5-2.0|=0.5$ m

Object 2:

$|3.8-3.0|=0.8$ m

Object 3:

$|5.0-4.0|=1.0$ m

## (b) Average Absolute Error

Stereo Vision:

30+0+0 =0 m

SfM:

30.5+0.8+1.0 =32.3 =0.77 m

## (c) Comparison

| Method | Average Absolute Error |
|---|---|
| Stereo Vision | 0 m |
| SfM | 0.77 m |

## Final Answer

Stereo Vision provides more accurate depth estimation because its average absolute error is 0 m, meaning its estimated depths exactly match the actual depths.

SfM has an average error of 0.77 m, so its estimates are less accurate for this dataset.

Therefore: Stereo Vision is the better method for depth estimation in this case.